Functional equations

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1 Introduction

In a functional equation, one tries to determine all functions satisfying a certain equation. One may know some functions like trigonometric functions, polynomials, exponential functions etc., but one cannot list all functions as such. The definition is very general.

Definition 1.1. *A function $f\colon A \to B : a \mapsto f(a)$ is any mapping that assigns to every element $a \in A$ an element $f(a) = b \in B$.*

For finite sets, there are $|B|^{|A|}$ many such functions. In that case, one could list all the functions by hand.
Nevertheless, this may soon be infeasible.

Example 1.2. *Deteremine all $f : \frac{\mathbb{Z}}{5\mathbb{Z}} \to \frac{\mathbb{Z}}{5\mathbb{Z}} : x \mapsto f(x)$ satisfying $f(a+b) = f(a)^2 + f(b^2)$ for every $a, b \in \frac{\mathbb{Z}}{5\mathbb{Z}}$.*

There are 5^5 many possibilities to verify if checking all of them.
Nevertheless, this one is easily solved by using some basic properties and choices of (a, b).

- Determine $f(0)$.
- Restrict the possibilities of outcomes for $f(a)$.
- Conclude.

In particular, we notice that one given equation can reduce the set of functions satisfying it a lot.
When we know the function has also an additional property or belongs to a specific class of functions, there are even more tools and elementary properties to use. E.g.

Observation 1.3. *If P is a polynomial with integer coefficients, i.e., $P(X) \in \mathbb{Z}[X]$, then $a - b \mid P(a) - P(b)$ for every $a, b \in \mathbb{Z}$.*

Theorem 1.4. *Every polynomial can be linearly over $\mathbb{C}$. Every polynomial in $\mathbb{R}[X]$ is the product of degree one and degree two polynomials over $\mathbb{R}$.*

Theorem 1.5. *(Rolle) If $f : [a, b] \to \mathbb{R}$ (where $b > a$) is a differentiable function satisfying $f(a) = f(b)$, there exists $a < c < b$ for which $f'(c) = 0$.*

2 Summary ideas

- Substitutions
- Induction
- Prove claims about injectivity/ surjectivity (and thus bijectivity)
- Finding zeros or fixed points
- Use properties if given; continuity/ polynomials
- Solve for a dense subset of the domain
- Recurrent relations; if $f(f(n)) = af(n)+b$, this can be repeated to express $f^{(k)}(n)$ as a linear combination of $f(n)$ and 1.
- Reduce to known equations; Cauchy's equation:

$$f(x+y) = f(x) + f(y) \ \forall \ x, \ y \in S$$

  If $S = \mathbb{Q}$, $f(x) = cx$.
  If $S = \mathbb{Q}[\sqrt{2}]$, $f(a + b\sqrt{2}) = ac + bd$ $(a, b \in \mathbb{Q}, \ c, d \in \mathbb{Q}[\sqrt{2}]$.
  If $S = \mathbb{R}$, many solutions, determined by base of $\mathbb{R}$ over $\mathbb{Q}$. With e.g. continuity or monotonicity, we know $f(x) = cx$.

- Write f as the sum of an even - and an odd function
- Difference with expected solution (if unique); if f_0 is the unique solution, substitute $f(x) = f_0(x) + g(x)$ and prove that g is the zero function.
- W.l.o.g.'s by invariant under operation; if $f(x) + c$ or $cf(x)$ are solutions whenever f is, we can choose the value of $f(0)$ or $f(1)$

Warning: Determining all functions, implies you prove **both** certain functions satisfy the properties, and other functions do not. So also check that the final subset of functions you obtained all satisfy the question.

3 Exercices

warm-up 1 Find all $f : \mathbb{R} \to \mathbb{R}$ such that $f(x+y) = f(x) + y$ for every $x, y \in \mathbb{R}$.

NT There exists a polynomial $P(X) \in \mathbb{Z}[X]$ such that $P(0) = 1, P(2) = 3$ and $P(4) = 9$?

Romania '86 Let $f, g\colon \mathbb{N} \to \mathbb{N}$ with f surjective and g injective, and $f(n) \geq g(n)$ for every $n \in \mathbb{N}$. Prove that $f(n) = g(n)$ for every $n \in \mathbb{N}$.

IMC '99 There exists a bijection $f\colon \mathbb{N} \to \mathbb{N} : n \mapsto f(n)$ for which $\sum_{n=1}^{\infty} \frac{f(n)}{n^2}$ is finite?

IMC '99 Find all strictly monotonic functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f^2(x) \geq f(x+y)(f(x)+y) \; \forall x, y > 0.$$

Dutch TST '20 Determine all polynomials $P(x)$ with real coefficients that apply $P(x^2) + 2P(x) = P(x)^2 + 2$.

FMO '22 Find all $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(m+n)+f(m)f(n) = n^2 f(m)+m^2 f(n)+(m+n)^2-m^2n^2, \quad \forall m, n \in \mathbb{Z}.$$

Euler '25 Find all functions $f\colon \mathbb{Q}[\sqrt{2}] \to \mathbb{Q}[\sqrt{2}]$ for which $f(x+y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$ for all $x, y \in \mathbb{Q}[\sqrt{2}]$.

4 Hints

- Can you guess the functions satisfying it? Can you prove it has roughly this form?
- Do you know some structure of P by knowing $P(0)$?
- Assume the contrary. Consider the set $A = \{n : f(n) \neq g(n)\}$.
- What do you know about the values $f(x)$ for $n+1 \leq x \leq 3n$? Alternative: What do you know about $\sum_{n=1}^{k} \frac{f(n)}{n^2}$?
- Try to prove that $f(x)$ is sufficiently fast decreasing to obtain a contradiction.
- Let $Q(x) = P(x) - 1$.
- Note that $f(n) = f(n) - n^2$ satisfies

$$g(m+n) + g(m)g(n) = 0, \quad \forall m, n.$$

- It is sufficient to know $f(1)$ and $f(\sqrt{2})$.

5 Solutions and solution sketches

Problem 5.1 (warm-up). *Find all $f : \mathbb{R} \to \mathbb{R}$ such that $f(x+y) = f(x) + y$ for every $x, y \in \mathbb{R}$.*

Proof. Choose $x = 0$. Then $f(y) = y + f(0)$. Now since every function $f(x) = x + c$ works, these are exactly the solutions. $\square$

Problem 5.2 (NT). *There exists a polynomial $P(X) \in \mathbb{Z}[X]$ such that $P(0) = 1, P(2) = 3$ and $P(4) = 9$?*

Proof. No. Note that $P(X) = XQ(X) + 1$ for some $Q \in \mathbb{Z}[X]$. Here $Q(2) = 1$ and $Q(4) = 2$. So $4 - 2 \nmid Q(4) - Q(2)$, from which we conclude. $\square$

Problem 5.3 (Romania '86). *Let $f, g \colon \mathbb{N} \to \mathbb{N}$ with f surjective and g injective, and $f(n) \geq g(n)$ for every $n \in \mathbb{N}$. Prove that $f(n) = g(n)$ for every $n \in \mathbb{N}$.*

Proof. Assume not. Consider the set $A = \{n : f(n) \neq g(n)\}$. Let $B = \{f(n) \mid n \in A\}$ and $C = \{g(n) \mid n \in A\}$. Then by definition, $\min B > \min C$ on the one hand, but $B = C$ on the other hand. $\square$

Problem 5.4 (IMC '99). *There exists a bijection $f \colon \mathbb{N} \to \mathbb{N} : n \mapsto f(n)$ for which $\sum_{n=1}^{\infty} \frac{f(n)}{n^2}$ is finite?*

Proof. Decrease values above k if necessary, and conclude that $\sum_{n=1}^{k} \frac{f(n)}{n^2} \geq \sum_{n=1}^{k} \frac{1}{k}$. The latter goes to infinity, from which the conclusion follows. $\square$

Problem 5.5 (IMC '99). *Find all strictly monotonic functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that*

$$f^2(x) \geq f(x+y)(f(x)+y) \ \forall x, y > 0.$$

Proof. Note that f is decreasing and that the inequality is equivalent with $f(x+y) \leq f(x) - \frac{f(x)y}{f(x)+y}$. Choose $y = \frac{1}{n} < f(x+1)$. Then $f(x+ky) < f(x+(k-1)y) - \frac{y}{2}$ for every $1 \leq k \leq n-1$ and thus $f(x+1) \leq f(x) - 1/2$. Hence $f(x+1+2n) < 0$, contradiction. $\square$

Problem 5.6 (Dutch TST '20). *Determine all polynomials $P(x)$ with real coefficients that apply $P(x^2) + 2P(x) = P(x)^2 + 2$.*

Proof. Let $Q(x) = P(x) - 1$, then $Q(x^2) = Q(x)^2$. If Q is not a constant function and has a (possibly complex) root r different from 0 and 1, then so is $\sqrt{r}$, leading to infinitely many roots. If $Q(1) = 0$, also $Q(-1) = 0$ and thus a contradiction is obtained. Hence $Q(x) = cx^n$, and $c \in \{0, 1\}$ (since c2 = c), and these also work. We conclude that $P(X) = 1$ or $P(X) = x^n + 1$ for some $n \geq 0$. $\square$

Problem 5.7 (FMO '22). *Find all $f : \mathbb{Z} \to \mathbb{Z}$ such that*

$$f(m+n) + f(m)f(n) = n^2 f(m) + m^2 f(n) + (m+n)^2 - m^2n^2, \quad \forall m, n \in \mathbb{Z}.$$

Proof. The function $g(n) = f(n) - n^2$ satisfies $g(m+n) + g(m)g(n) = 0, \quad \forall m, n$. In particular $g(n)(1 + g(0)) = 0$. If $g(0) \neq -1$, then $g \equiv 0$.
If $g(0) = -1$, choosing $m = -n$ results in $g(m) = g(n) = \pm 1$. Here $m = n$ implies that $g(2n) = -1$. Finally, we have $g(2n+1) = g(1) \in \{-1, 1\}$.
The solutions are $f(n) = n^2$, $f(n) = n^2 - 1$ and $f(n) = n^2 - (-1)^n$. Finally, one checks those three functions. □

Problem 5.8 (Euler '25). *Find all functions $f \colon \mathbb{Q}[\sqrt{2}] \to \mathbb{Q}[\sqrt{2}]$ for which $f(x+y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$ for all $x, y \in \mathbb{Q}[\sqrt{2}]$.*

Proof. We know by Cauchy's equation (classical form) that $f(a+b\sqrt{2}) = af(1) + bf(\sqrt{2})$. Now $f(2) = f(\sqrt{2})^2$ and $2f(2) = f(4) = f(2)^2$. This implies that $f(2) \in \{0, 2\}$. Furthermore, in the second case $f(\sqrt{2}) = \pm\sqrt{2}$.
We conclude that the functional equation has three solutions; $f(z) = 0$, $f(z) = z$ and $f(z) = \overline{z}$ for all $z \in \mathbb{Q}[\sqrt{2}]$. Here $\overline{a + b\sqrt{2}} = a - b\sqrt{2}$. The first two trivially satisfy the equalities. The third one is also checked by direct verification. □